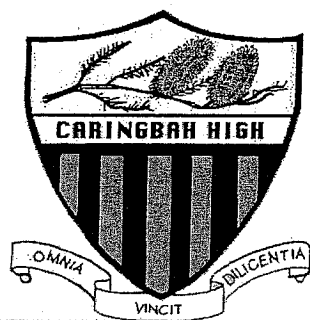


Student Name:

2004
TRIAL HIGHER SCHOOL CERTIFICATE

MATHEMATICS
Extension 1



General Instructions

Reading Time: 5 minutes
Working Time: 2 hours

- Attempt all questions
- Start each question on a new page
- Each question is of equal value
- Show all necessary working.
- Marks may be deducted for careless work or incomplete solutions
- Standard integrals are printed on the last page
- Board-approved calculators may be used

Question One

Marks

a) Find the remainder when $P(x) = x^3 - 4x + 2$ is divided by $x - 1$.

1

b) Find $\int x e^{x^2} dx$

1

c) Solve the inequality $\frac{2x-3}{x} \leq 4$

3

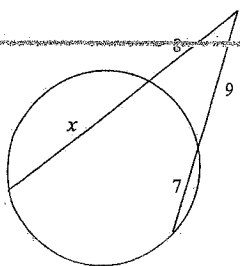
d) For the points $A(3, -5)$ and $B(-4, 2)$, find the coordinates of the point P which divides the interval AB externally in the ratio 2:1.

2

e) Solve the equation $x - 1 = \sqrt{x + 1}$.

3

f)



Not to scale

Calculate the value of x ,
giving a reason for your answer.

2

Question Two (Start a new page)

Marks

a) Find $\int \frac{1}{x \log_e x} dx$ using the substitution $u = \log_e x$.

2

b) The cubic equation $x^3 - 4x^2 + x + 1 = 0$ has a root near $x = 0.7$

2

Use one application of Newton's Method to find a better approximation, giving your answer to 2 decimal places.

c) For the function $f(x) = 2 \sin^{-1}\left(\frac{x}{2}\right)$

i) Find $f(2)$

1

ii) State the domain and range of this function.

1

iii) Neatly sketch $y = f(x)$.

2

d) A parabola is defined by the parametric equations

$$\begin{aligned} x &= 12t \\ y &= 6t^2 + 3 \end{aligned}$$

i) Find $\frac{dx}{dt}$ and $\frac{dy}{dt}$.

1

ii) If Q is the point where $t = -1$, find

α) the coordinates of Q .

1

β) the slope of the tangent to the curve at Q .

1

γ) the equation of the tangent at Q .

1

Question Three (Start a new page)

Marks

a) Evaluate $\lim_{x \rightarrow 1} \frac{x^3 - 1}{x^2 - 1}$

2

b) Evaluate $\int_{-3}^0 \frac{x}{\sqrt{x+4}} dx$ using the substitution $u^2 = x+4$, where $u > 0$.

3

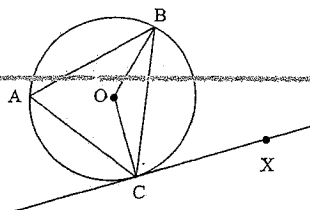
c) Prove, by Mathematical Induction, that for $n \geq 1$

4

$$\frac{1}{1 \times 4} + \frac{1}{4 \times 7} + \frac{1}{7 \times 10} + \dots + \frac{1}{(3n-2) \times (3n+1)} = \frac{n}{3n+1}$$

d)

CX is tangent to the circle centre O. Let $\angle CAB = \alpha$.



i) Copy the diagram to your answer sheet.

ii) Find with reasons $\angle COB$ in terms of α .

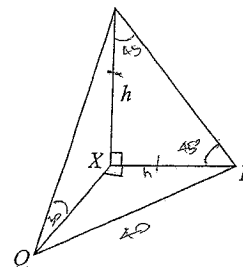
iii) Find with reasons $\angle OCB$ in terms of α .

iv) Hence show that $\angle BCX = \angle BAC$.

Question Four (Start a new page)

Marks

- a) A vertical tower with base X and height h metres stands on horizontal ground. From a point P on the ground due east of the tower the angle of elevation of the top of the tower is 45° and from a point Q on the ground due south of the tower the angle of elevation of the top of the tower is 30° . If distance PQ is 40 metres:



i) Find the length of QX in terms of h .

1

ii) Hence find the height of the tower.

2

b) Find $\int \frac{1}{4+x^2} dx$

1

- c) A sphere is being heated so that its surface area is increasing at a constant rate of $15 \text{ mm}^2/\text{s}$. Find the rate of increase of the volume of the sphere when the radius is 5 mm . (You are given that $V = \frac{4}{3}\pi r^3$ and $S.A. = 4\pi r^2$).

3

- d) Corn cobs are cooked by immersing them in boiling water. On being removed, a corn cob cools in the air according to the equation $\frac{dT}{dt} = -k(T - B)$, where t is time in minutes, T is temperature in degrees celsius ($^\circ\text{C}$), B is the temperature of the surrounding air and k is a positive constant.

i) Show that $T = B + Ae^{-kt}$ is a solution of the above equation where A is a constant.

1

ii) If the temperature of the boiling water is 100°C and the surrounding air is a constant 25°C , find the value of A and the value of k (correct to 4 decimal places) if a corn cob cools to 70°C in 3 minutes.

2

iii) How long (to the nearest minute) will it take for the corn to cool to 50°C ?

2

Question Five (Start a new page)

Marks

a) If $y = \sin^{-1}(3x-2)$, find $\frac{dy}{dx}$. 2

b) If p, q , and r are the roots of the cubic equation $x^3 + 2x^2 + 3x + 5 = 0$, find the value of:

i) $p + q + r$. 1

ii) $p^{-1} + q^{-1} + r^{-1}$. 3

c) $A(t, e^t)$ and $B(-t, e^{-t})$ are points on the curve $y = e^x$, where $t > 0$.

The tangents at A and B form an angle of 45° .

i) Prove that $e^t - \frac{1}{e^t} = 2$. 3

ii) Hence by solving the equation in part (i) find the coordinates of A in **exact form**. 3

Question Six(Start a new page)

Marks

a) Consider the function $f(x) = \frac{x+1}{x^2+3}$

i) Find the points where the curve crosses the x -axis and the y -axis. 1

ii) Find the coordinates of any stationary points on the curve $y = f(x)$ and, without finding the second derivative, determine their nature. 3

iii) Describe the behaviour of $y = f(x)$ for large positive and large negative values of x . 1

iv) Using an **appropriate scale** neatly sketch $y = f(x)$ showing all important information. 2

b) The points $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ lie on the parabola $x^2 = 4ay$.

[You are given that the equation of the chord PQ is $2y = (p+q)x - 2apq$]

i) If the chord PQ passes through $(2a, 0)$ show that $pq = p+q$. 1

ii) Hence, find the locus of M , the midpoint of PQ . 4

Question Seven(Start a new page)

- a) An urn contains W white balls and B black balls.

If the probability of selecting 2 white balls in succession at random is $\frac{1}{3}$ and

the probability of selecting 3 white balls in succession at random is $\frac{1}{6}$, find

the number of white balls in the urn.

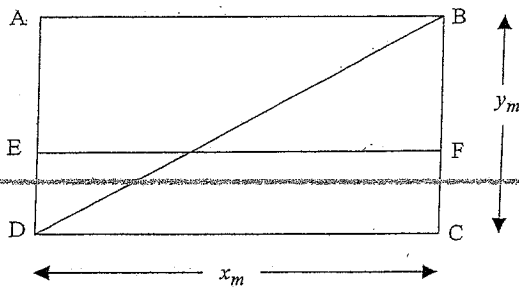
[You must justify your answer to gain full marks]

- b) A particle is travelling in a straight line executing Simple Harmonic Motion about O according to the equation $x = a \cos nt$.

i) Show that the velocity v and the displacement x of the particle at any time t are related by the equation $v^2 = n^2(a^2 - x^2)$.

ii) Hence using part (i) show that the acceleration of the particle can be written as $\ddot{x} = -n^2x$.

- c) A rectangle ABCD with sides of length x metres and y metres has an area of $9m^2$. Two metal construction strips, one a diagonal BD and the other EF parallel to the sides AB and CD are required to strengthen the rectangle.



- i) Show that the total length L of both strips is given by

$$L = x + \frac{\sqrt{x^4 + 81}}{x} \text{ metres.}$$

- ii) Find the value of x that will minimise the total length L of the strips, writing your answer in the form $x = a^{b/c}$, where a, b and c are integers.

[You do not need to justify that your value of x gives a minimum L]

END OF EXAM

Marks

3

2

1

2

4

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0; \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

Question One

a) Using the remainder theorem find $P(1) = -1$.

b) $\frac{1}{2}e^{x^2} + c$

$$\begin{aligned} \text{c) } x^2 \left(\frac{2x-3}{x} \right) &\leq 4x^2 \\ x(2x-3) - 4x^2 &\leq 0 \\ -x(2x+3) &\leq 0 \end{aligned}$$

$$\therefore x \leq -\frac{3}{2} \text{ and } x > 0 \text{ as } x \neq 0.$$

d) $m:n = 2:-1$

$$\therefore x = \frac{2 \times -4 + -1 \times 3}{2 + (-1)} = -11, y = \frac{2 \times 2 + -1 \times 3}{2 + (-1)} = 9$$

 $\therefore P$ has coordinates $(-11, 9)$

e) $(x-1)^2 = x+1$

$$x^2 - 2x + 1 = x + 1$$

$$x^2 - 3x = 0 \Rightarrow x(x-3) = 0$$

$$\therefore x = 0 \text{ or } x = 3$$

but as x must ≥ 1 , then $x = 3$ only.

$$\begin{aligned} \text{f) } 8 \times (x+8) &= 9 \times 16 \quad [\text{Product of intersecting secants}] \\ 8x + 64 &= 144 \Rightarrow x = 10. \end{aligned}$$

Question Two

a) $u = \ln x \Rightarrow du = \frac{1}{x} dx$

$$\begin{aligned} \therefore I &= \int \frac{1}{u} du = \ln(u) \\ &= \ln(\ln x) + c \end{aligned}$$

b) Let $P(x) = x^3 - 4x^2 + x + 1$

$$\therefore P'(x) = 3x^2 - 8x + 1$$

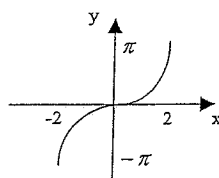
$$\text{Now with } x_1 = 0.7, x_2 = 0.7 - \frac{P(0.7)}{P'(0.7)} = 0.73 \text{ (2dp)}$$

c) i) π

ii) Domain: $-2 \leq x \leq 2$

Range: $-\pi \leq y \leq \pi$

iii)



d) i) $\frac{dx}{dt} = 12$ and $\frac{dy}{dt} = 12t$

ii) $\alpha) Q(-12, 9)$

$$\beta) \frac{dy}{dx} = \frac{dy}{dt} \times \frac{dt}{dx} \text{ using the chain rule}$$

$$= 12t \times \frac{1}{12} = -1 \text{ when } t = -1.$$

$$\gamma) y - 9 = -1(x + 12)$$

$$\therefore y = -x - 3.$$

Question Three

a) $\lim_{x \rightarrow 1} \frac{(x-1)(x^2+x+1)}{(x-1)(x+1)}$

$$= \lim_{x \rightarrow 1} \frac{x^2+x+1}{x+1} = \frac{1+1+1}{1+1} = \frac{3}{2}$$

b) $u^2 = x+4 \Rightarrow 2u du = dx$ and $u = \sqrt{x+4}$

Also, when $x = -3$, $u = 1$ and when $x = 0$, $u = 2$.

$$\therefore I = \int_1^2 \frac{u^2 - 4}{u} \times 2u du$$

$$= 2 \int_1^2 (u - \frac{4}{u}) du$$

$$= 2 \left[\frac{u^2}{2} - 4 \ln u \right]_1^2$$

$$= 2 \left[\left(\frac{8}{2} - 8 \ln 2 \right) - \left(\frac{1}{2} - 4 \ln 1 \right) \right] = -10 \ln 2$$

c) When $n = 1$, LHS = $\frac{1}{4}$, RHS = $\frac{1}{3 \times 1 + 1} = \frac{1}{4}$

 $\therefore$ true for $n = 1$.
Assume true for $n = k$, i.e.

$$\frac{1}{1 \times 4} + \frac{1}{4 \times 7} + \frac{1}{7 \times 10} + \dots + \frac{1}{(3k-2) \times (3k+1)} = \frac{k}{3k+1}$$

Now prove true for $n = k+1$, i.e. prove that

$$S_k + T_{k+1} = S_{k+1}.$$

$$S_k = \frac{k}{3k+1}, S_{k+1} = \frac{k+1}{3k+4}, T_{k+1} = \frac{1}{(3k+1)(3k+4)}$$

$$\therefore S_k + T_{k+1} = \frac{k}{3k+1} + \frac{1}{(3k+1)(3k+4)}$$

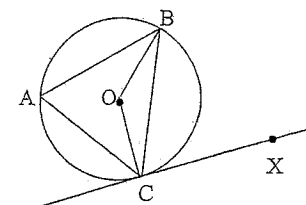
$$= \frac{k(3k+4) + 1}{(3k+1)(3k+4)}$$

$$= \frac{3k^2 + 4k + 1}{(3k+1)(3k+4)}$$

$$\begin{aligned} &= \frac{(3k+1)(k+1)}{(3k+1)(3k+4)} = \frac{k+1}{3k+4} \\ &= S_{k+1} \end{aligned}$$

 $\therefore$ If true for $n = k$ then true for $n = k+1$
and since true for $n = 1$, then true for
 $n = 2$ and so on for all integers n .

d)

i) $\angle COB = 2\alpha$ ($\angle$ at centre = twice $\angle$ at circumference)ii) $\angle OCB = 90 - \alpha$ (base $\angle$'s of isos Δ = and $\angle$ sum of Δ)iii) $\angle BCX = \angle OCX - \angle OCB$
 $= 90 - (90 - \alpha)$ (rt. $\angle$, tangent $\perp$ to radius)

$$= \alpha$$

$$= \angle BAC.$$

Question Four

a) i) $\tan 30^\circ = \frac{h}{OQ} \Rightarrow OQ = \sqrt{3}h$

ii) Similarly $\tan 45^\circ = \frac{h}{PX} \Rightarrow PX = h$

$$\therefore \text{Using } \Delta PXQ: 40^2 = (\sqrt{3}h)^2 + h^2$$

$$1600 = 4h^2 \Rightarrow h = 20m$$

b) $\frac{1}{2} \tan^{-1} \left(\frac{x}{2} \right) + c$

Question Four (cont)

$$c) \frac{dA}{dt} = 15, \frac{dA}{dr} = 8\pi r, \frac{dV}{dr} = 4\pi r^2, \frac{dV}{dt} = ?$$

Need to use the chain rule twice:

$$\frac{dA}{dt} = \frac{dA}{dr} \times \frac{dr}{dt} \Rightarrow 15 = 8\pi r \times \frac{dr}{dt} \Rightarrow \frac{dr}{dt} = \frac{15}{8\pi r}$$

$$\text{Also } \frac{dV}{dt} = \frac{dV}{dr} \times \frac{dr}{dt} = 4\pi r^2 \times \frac{15}{8\pi r}$$

$$\therefore \frac{dV}{dt} = \frac{15r}{2} \Rightarrow 37.5\text{mm}^3/\text{s} \text{ when } r=5.$$

$$d) i) LHS = \frac{dT}{dt} = -kAe^{-kt}$$

$$RHS = -k(T-B) = -k(B + Ae^{-kt} - B) = -kAe^{-kt} = LHS$$

$$ii) \text{ When } t=0, T=100^\circ \text{ and when } t=3, T=70^\circ.$$

$$\text{Now } T = 25 + Ae^{-kt}$$

$$\therefore 100 = 25 + Ae^0 \Rightarrow A = 75.$$

$$\text{Also } 70 = 25 + 75e^{-3k}$$

$$e^{-3k} = \frac{45}{75} \Rightarrow -3k = \ln\left(\frac{3}{5}\right) \Rightarrow k \approx 0.1703$$

$$iii) 50 = 25 + 75e^{-0.1703t} \Rightarrow \frac{1}{3} = e^{-0.1703t}$$

$$\therefore -0.1703t = \ln\left(\frac{1}{3}\right) \Rightarrow t \approx 6 \text{ minutes.}$$

Question Five

$$a) \frac{dy}{dx} = \frac{3}{\sqrt{1-(3x-2)^2}}$$

$$b) i) p+q+r = -\frac{b}{a} = -2$$

$$ii) pq+pr+qr = \frac{c}{a} = 3 \text{ and } pqr = -\frac{d}{a} = -5$$

$$\therefore p^{-1} + q^{-1} + r^{-1} = \frac{1}{p} + \frac{1}{q} + \frac{1}{r} = \frac{pq+pr+qr}{pqr} = -\frac{3}{5}$$

$$c) y = e^x \Rightarrow \frac{dy}{dx} = e^x$$

$$\therefore \text{ at } A: x=t \Rightarrow m_1 = e^t$$

$$\text{and at } B: x=-t \Rightarrow m_2 = e^{-t}$$

$$\therefore \tan 45^\circ = \left| \frac{e^t - e^{-t}}{1 + e^t \times e^{-t}} \right|$$

$$\therefore 1 = \frac{e^t - e^{-t}}{1 + e^0} \Rightarrow 2 = e^t - e^{-t}$$

$$\text{Hence } 2 = e^t - \frac{1}{e^t} \quad **$$

$$ii) (e^t)^2 - 2e^t - 1 = 0 \text{ using } **$$

$$\therefore e^t = \frac{2 \pm \sqrt{(-2)^2 - 4 \times 1 \times -1}}{2 \times 1}$$

$$= \frac{2 \pm \sqrt{8}}{2} = \frac{2 \pm 2\sqrt{2}}{2} = 1 \pm \sqrt{2}$$

$$\text{but as } e^t > 0, \text{ then } e^t = 1 + \sqrt{2} \Rightarrow t = \ln(1 + \sqrt{2})$$

$$\therefore A \text{ has coordinates } \left[\ln(1 + \sqrt{2}), (1 + \sqrt{2}) \right]$$

Question Six

$$a) i) \text{ At } x=0 \Rightarrow y = \frac{1}{3} \therefore y\text{-intercept is } \left(0, \frac{1}{3}\right)$$

$$\text{At } y=0 \Rightarrow x=-1 \therefore x\text{-intercept is } (-1, 0).$$

$$ii) \frac{dy}{dx} = \frac{(x^2+3) \times 1 - (x+1) \times 2x}{(x^2+3)^2} = \frac{3-2x-x^2}{(x^2+3)^2}$$

$$\text{For stationary points } \frac{dy}{dx} = 0$$

$$\therefore x^2 + 2x - 3 = 0 \Rightarrow (x+3)(x-1) = 0$$

$$\therefore x = -3 \text{ or } x = 1$$

$$\text{When } x = -3, y = -\frac{1}{6}$$

and

x	-3^-		-3^+
y'	$-ve$	0	$+ve$

$$\therefore \left(-3, -\frac{1}{6}\right) \text{ is a minimum.}$$

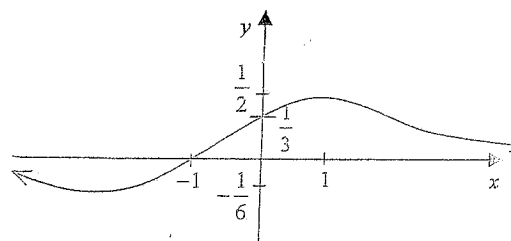
$$\text{When } x = 1, y = \frac{1}{2} \text{ and}$$

x	1^-		1^+
y'	$+ve$	0	$-ve$

$$\therefore \left(1, \frac{1}{2}\right) \text{ is a maximum.}$$

$$iii) \text{ As } x \rightarrow \infty, f(x) \rightarrow 0 \text{ from above.}$$

$$\text{and as } x \rightarrow -\infty, f(x) \rightarrow 0 \text{ from below.}$$



$$6b) i) \text{ Substitute the point } (2a, 0) \text{ into the equation}$$

$$2y = (p+q)x - 2apq \text{ to obtain}$$

$$0 = (p+q) \times 2a - 2apq$$

$$\therefore 0 = 2ap + 2aq - 2apq$$

$$0 = p + q - pq \Rightarrow pq = p + q$$

$$ii) \text{ The midpoint } M \text{ of the chord } PQ \text{ has}$$

$$\text{coordinates } M \left(a(p+q), a \left(\frac{p^2+q^2}{2} \right) \right)$$

$$\therefore x^2 = a^2(p+q)^2$$

$$= a^2(p^2 + q^2) + 2a^2pq$$

$$= 2a \times a \left(\frac{p^2+q^2}{2} \right) + 2a^2pq$$

$$= 2ay + 2a \times a(p+q) \text{ ----- using (i)}$$

$$= 2ay + 2ax$$

Hence the locus of M is given by the equation

$$x^2 = 2a(y+x)$$

Question Seven

$$a) \frac{W}{W+B} \times \frac{W-1}{(W-1)+B} = \frac{1}{3} \quad \text{----- (1)}$$

$$\frac{W}{W+B} \times \frac{W-1}{(W-1)+B} \times \frac{W-2}{(W-2)+B} = \frac{1}{6}$$

$$\therefore \frac{1}{3} \times \frac{W-2}{(W-2)+B} = \frac{1}{6} \Rightarrow \frac{W-2}{(W-2)+B} = \frac{1}{2}$$

$$\therefore 2W-4 = W-2+B \Rightarrow B = W-2 \quad \text{--(2)}$$

Substitute (2) into (1) to obtain

$$\frac{W}{2W-2} \times \frac{W-1}{2W-3} = \frac{1}{3}$$

$$\frac{W^2 - W}{4W^2 - 10W + 6} = \frac{1}{3}$$

$$3W^2 - 3W = 4W^2 - 10W + 6$$

$$W^2 - 7W + 6 = 0$$

$$(W-1)(W-6) = 0$$

$$\therefore W = 1 \text{ or } W = 6$$

But $W = 1$ is trivial hence there are six white balls.

$$b) i) x = a \cos nt$$

$$\therefore v = -an \sin nt$$

$$\begin{aligned} \text{Hence } v^2 &= a^2 n^2 \sin^2 nt \\ &= a^2 n^2 (1 - \cos^2 nt) \\ &= a^2 n^2 - n^2 (a \cos nt)^2 \\ &= a^2 n^2 - n^2 x^2 \\ &= n^2 (a^2 - x^2) \end{aligned}$$

$$\begin{aligned} ii) \ddot{x} &= \frac{d}{dx} \left(\frac{1}{2} v^2 \right) \\ &= \frac{1}{2} n^2 \frac{d}{dx} [a^2 - x^2] \\ &= \frac{1}{2} n^2 \times -2x \\ &= -n^2 x \end{aligned}$$

$$c) i) xy = 9 \Rightarrow y = \frac{9}{x} \quad \text{----- (1)}$$

$$\text{Also } L = DC + DB = x + \sqrt{x^2 + y^2}$$

$$\therefore L = x + \sqrt{x^2 + \left(\frac{9}{x}\right)^2} \quad \text{using (1)}$$

$$= x + \sqrt{x^2 + \frac{81}{x^2}} = x + \sqrt{\frac{x^4 + 81}{x^2}}$$

$$\therefore L = x + \frac{\sqrt{x^4 + 81}}{x}$$

$$ii) \frac{dL}{dx} = 1 + \frac{x \times \frac{1}{2} (x^4 + 81)^{-\frac{1}{2}} \times 4x^3 - \sqrt{x^4 + 81} \times 1}{x^2}$$

$$= 1 + \frac{\frac{2x^4}{\sqrt{x^4 + 81}} - \sqrt{x^4 + 81}}{x^2}$$

$$= 1 + \frac{2x^4 - (x^4 + 81)}{x^2 \sqrt{x^4 + 81}}$$

$$= 1 + \frac{x^4 - 81}{x^2 \sqrt{x^4 + 81}}$$

$$\text{For a minimum } \frac{dL}{dx} = 0 \Rightarrow 1 = \frac{81 - x^4}{x^2 \sqrt{x^4 + 81}}$$

$$\therefore x^2 \sqrt{x^4 + 81} = 81 - x^4 \text{ and square both sides}$$

$$x^4 (x^4 + 81) = 6561 - 162x^4 + x^8$$

$$x^8 + 81x^4 = 6561 - 162x^4 + x^8$$

$$6561 = 243x^4$$

$$\therefore x^4 = 27$$

$$\text{Hence } x = 27^{\frac{1}{4}} \text{ or } x = 3^{\frac{3}{4}}$$